Devinder Singh Saggu

Ottawa, Ontario – K0A 2Z0

Phone: +1 (613) 501-2158 | **Email**: saggudevindersingh@gmail.com | **LinkedIn**: linkedin.com/in/devinder-singh-saggu

Professional Summary

Proven technical leader with a track record of driving system design, authoring architectural specs, and guiding engineering teams through complex SaaS and microservices deliverables. Experienced in conducting code reviews, establishing quality standards, unblocking developers, and driving Agile team execution to accelerate release velocity.

Hands-on experience with CI/CD pipelines, GitHub Actions, Docker, Kubernetes, and Azure cloud infrastructure, along with AI-assisted engineering workflows and **Spec-Driven Development (SDD)** using tools such as Cursor, Claude CLI, GitHub Copilot, and ChatGPT to accelerate development, testing, and code quality. Proven ability to collaborate across teams, contribute to architecture decisions, mentor developers, and deliver high-quality enterprise applications using modern software engineering best practices.

Professional Experience

Senior Full Stack Developer | BASF — May. 2026 – Current

- Led the development of a **Spring Boot** based **SaaS application** for managing seed allocation and distribution to agricultural retailers across multiple regions, enabling scalable data processing, regional insights, and improved system reliability.
- Designed and developed responsive frontend data tables using **React**, integrating them with **Node.js** and **Spring Boot** APIs documented via **OpenAPI / Swagger** to ensure clear API contracts and efficient end-to-end data processing.
- Designed and implemented database schemas using **Flyway** and mapped database entities with **Hibernate/JPA**, ensuring consistent and maintainable data persistence.
- Implemented Single Sign-On (SSO) and authentication workflows using **OAuth 2.0** and **OpenID Connect (OIDC)** via Azure Entra ID, enhancing Cortex application security and access management across role-based user-groups.
- **Drove cross-functional engineering practices across sprint teams**, enforcing SonarQube quality gates, automating GitHub Actions CI/CD pipelines on Azure, and **increasing test coverage to accelerate production release cadence**.
- Developed and maintained **automated frontend test cases using Playwright and Jest** for React components, covering regional scenarios including **Supply, West, and East**, improving test coverage and ensuring reliable application functionality across different data configurations.

Full Stack Developer | Ross Video — Nov. 2019 – May 2026

- Designed and delivered a **Spring Boot** based application to visualize switcher performance, integrating secure REST APIs, MySQL and Grafana, reducing manual analysis effort by over 80% to improve engineering velocity.
- Designed and implemented distributed caching using **Hazelcast** within microservices architecture, improving system availability and fault tolerance while reducing inter-service latency by 50% in a large-scale distributed system.
- Designed and implemented a scalable publish/deploy feature in ProjectServer with an React-based GUI, enabling multi-client real-time communication using **concurrent processing** and **multithreading**.
- Implemented event-driven **WebSocket communication** between OverDrive and ProjectServer enabling real-time state synchronization in distributed systems without polling.
- Worked with cross-functional teams to enhance platform stability by containerizing OverDrive services using **Docker**, improving consistency and portability across environments.
- Led the design and delivery of an **internal mock license server**, eliminating third-party dependency and reducing regression testing cycle time by over 70%, **accelerating release** readiness.
- Migrated **Java libraries to Python libraries** integrated with Robot Framework, improving test coverage and accelerating identification of performance regressions.
- Leveraged **AI-assisted** tools (Cursor, Claude CLI) to accelerate development, improve test coverage, and enhance legacy code maintainability.
- Leveraged AWS **CloudWatch** and **CloudTrail** for centralized logging, monitoring, and audit analysis in OverDrive services, enabling faster issue diagnosis and improving system observability.
- Actively contributed to **system design**, **architectural discussions**, and **code reviews**, improving overall code quality and reducing technical debt.
- **Mentored** junior developers, conducted code reviews, and drove architectural decisions to improve code quality and system stability.
- Collaborated with QA, product owners, and **cross-functional teams** in an Agile/Scrum environment to deliver high-quality software.
- Performed sprint, company-wide and client-facing demos for the new features and system capabilities.

Software Developer | Nucleus Software — Aug. 2014 – Dec. 2017

Project - FinnOne NEO Customer Acquisition System (CAS)

- Designed and implemented validation and business logic for financial transactions in banking and capital market system, including Debt Burden Ratio (**DBR**) calculations for credit risk assessment, ensuring data integrity and adherence to regulatory and banking standards.
- Designed and developed new features for a **Spring MVC** based web-application, improving functionality and user-experience.

- Applied secure coding practices using **Spring Security** and input validation to ensure data integrity and compliance in financial systems.
- Developed and deployed **REST** and **SOAP**-based web service to calculate the eligible loan amount for customers.
- **Optimized SQL queries** using JProfiler analysis, enhancing application performance and throughput.
- Acted as **Scrum Master**, driving Agile best practices and ensuring successful sprint delivery using Jira for tracking and coordination.
- Redesigned the **RCU module** of the application by adding the new features and intensifying the front-end user interface.
- Wrote the **JUnit** test cases to ensure the correct functionality of the application and avoid troubleshoot problems.

Project - Credit Card web-application
- Developed and deployed **RESTful web services** to determine eligible clients, streamlining application processes.
- Integrated **Spring Security** to provide **role-based access control (RBAC)** and user-specific privileges.

Project - Client Support team
- Supported **HDFC, ICICI, and DCB bank clients** by resolving production issues efficiently.
- Implemented **custom modifications for financial reports**, enhancing client satisfaction and compliance.

Technical Skills

- **Programming Languages:** Java 8, Java 17, J2EE, Python, TypeScript, JavaScript, HTML, CSS
- **Frontend:** React, Node.js, Angular, JavaScript, TypeScript, jQuery, HTML, CSS
- **Frameworks & APIs:** Spring Framework**,** Spring Boot, Spring Security, JPA(Hibernate)
- **APIs and Communication:** REST APIs, WebSocket, gRPC
- **Databases:** SQL, PostgreSQL, MySQL, Oracle
- **DevOps & CI/CD:** Git, Jenkins, GitHub Actions, Docker, Relic, Kubernetes, Maven, Terraform/Iac
- **Cloud & Infrastructure:** Azure DevOps, Azure App Services
- **Event-Driven & Messaging:** ActiveMQ, RabbitMQ, Hazelcast
- **Testing & Quality:** JUnit (TDD), Mockito, ArchUnit, Robot Automation
- **Observability:** CloudWatch, Grafana, Metrics, Alerting
- **Developer Productivity Tools:** JIRA, SonarQube, GitHub Copilot, Cursor, Claude CLI, ChatGPT
- **Methodologies:** Agile/Scrum, SDLC

Education

Master of Computer Engineering (Data Science specialization) Jan. 2018 – Aug. 2019
Carleton University, Ottawa, Ontario

Bachelor of Technology in Electronics and Communication Engineering June 2010 – May 2014
Dr B R Ambedkar National Institute of Technology Jalandhar, Punjab, India

Certifications

- AWS Cloud Practitioner (Amazon Web Services Training and Certification)
- Structuring Machine Learning Projects (Certificate)
- Improving Deep Neural Networks: Hyperparameter tuning, Regularization, and Optimization (Certificate)
- Neural Networks and Deep Learning (Certificate)

Applied Projects

A Comparative Study of Machine Learning Techniques Jan. 2019 – Apr. 2019
Carleton University, Ottawa, ON
- Collected, Visualized and Preprocessed traffic data of Ottawa city maintained by the Transportation Services Department
- Designed accident prediction model following ***the CRISP-DM*** methodology
  Technologies and Techniques used: Tableau, MS Office, Rapid Miner, Python, Networks and Deep Learning

Research Assistant – Data Engineer
Carleton University, Ottawa, Ontario Nov. 2018 – Aug. 2019
- Collected, explored and visualized the financial data of all Municipalities, Ministries of different Provinces and Ministries of Federal of Canada
- Merged the collected financial data with another dataset to determine the salary patterns among different employees

High-performance e-commerce application Jan. 2018 – April 2018
Carleton University, Ottawa, ON
- Designed LQN diagram and sequence diagram to represent the different operations such as View Profile, Add/Update products to cart and Make Payments
- Successfully implemented multithreading, concurrency and spreading the load principle to get minimum response time and maximum throughput of the application